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EDUCATION

Singapore University of Technology and Design

B.Eng. Engineering Systems and Design | GPA: 4.16/5.00

- Specialisations: Financial Services; Business Analytics & Operations Research
- Awards: Best Project in Data Driven World; Top 3 in Design Thinking 3; Sustainability Design Prize

Singapore

Expected May 2028

National Taiwan University

Semester Exchange, SUTD Global Exchange Programme

- Coursework: Econometric Theory I (graduate), Stochastic Calculus, Causal ML, Machine Learning, Algorithms, Game Theory

Taipei, Taiwan

Aug – Dec 2026

COMPETITIONS & LEADERSHIP

IMC Prosperity 4

Finalist | Ranked 10th Singapore / #474 Global out of 18,000+ teams

- Built Python market-making, options-pricing, and relative-value strategies; designed a data-driven selection framework for Round 5's 50-product universe.

Singapore

Apr 2026

WorldQuant International Quant Championship 2026

Gold Tier | Stage 2: Top 22 Singapore / #1,376 Global

- Built market-neutral equity alphas across price, fundamental, and sentiment signals (top IS Sharpe 2.21; Fitness 2.30).
- Built a Python BRAIN API pipeline: expression canonicalizer, template-based generation, SQLite experiment tracking.

Singapore

Mar – Jun 2026

AlphaNova Competition 5 – Cross-Sectional Signal Forecasting

Quality Set qualifier | #140 pre-simulation leaderboard; live simulation ongoing

- Engineered Ridge ranking signals (z-scores, rank velocity); two passed overfitting and originality checks for prizes.
- Fixed a silent feature–target misalignment from MultiIndex column ordering via explicit (time, ticker) joins.

Remote

Jul – Oct 2026

SUTD MarketWatch Club

Treasurer

- Managed S\$10,000+ in club funds (budgeting, claims, event allocation) for fortnightly trading and macro workshops.

Singapore

Feb 2025 – Mar 2026

PROJECTS

Nonlinear Asset-Pricing Bubble Stabilization

MATLAB, Control Theory | Heuristics & Systems Theory (40.018)

- Modeled bubble dynamics in a heterogeneous-beliefs asset-pricing model (Brock & Hommes 1998; Schmitt et al. 2020); designed state-feedback and observer-based stabilizing controllers via Ackermann's formula, validated by simulation.

Jul – Aug 2026

Multi-Objective HIV Funding Allocation

NSGA-II, Simulated Annealing, Mixed-Integer Programming | Heuristics & Systems Theory (40.018)

- Formulated HIV funding allocation across African countries as a multi-objective optimization problem, comparing greedy, simulated annealing, and NSGA-II Pareto-front solutions against an exact MIP benchmark.

May – Jul 2026

Analytics Edge Data Competition – Discrete Choice Modeling

R, Conditional/Mixed Logit, XGBoost

- Built a conditional logit with covariate interactions and price-factor features, reaching a logit + XGBoost ensemble at 1.202 public log loss; resolved a model-matrix rank deficiency diagnosed via QR decomposition.

May – Aug 2026

Aggregate Planning – Aster Chemicals (Pulau Bukom Refinery)

Excel Solver, Linear Programming, Manufacturing & Service Operations

- Owned workforce planning in a 5-person team: built a Solver model comparing Chase, Level, Hybrid, and LP-Optimal plans, linearizing IF() constraints for Simplex; recommended Hybrid at S\$1.7M (18.9/20).

Jan – Jun 2026

CBS Synthetic Data Generator

Python, Streamlit, CTGAN | Industry-partner project, Data & Business Analytics

- Built a Streamlit synthetic-data tool for Credit Bureau Singapore with a CTGAN and schema-cleaning pipeline producing coherent credit-profile, delinquency, and share-of-wallet datasets for bank reporting.

Feb – May 2026

WORK EXPERIENCE

MolarData

Computer Vision Intern | via SUTD DGIT Programme at Zhejiang University

- Annotated and classified 2,000+ images in a 10-person team, producing training datasets for in-house CNN models.

Hangzhou, China

Oct – Dec 2025

CERTIFICATIONS & SKILLS

Certifications: CFA Level I Candidate (May 2027); SCI Life, Health, ILP, Rules & Ethics

Programming & Tools: Python, R, SQL, MATLAB, $\LaTeX$, Git, Excel (Solver), Bloomberg Terminal, Streamlit

Finance & Trading: Market-Making, Options Pricing, Algorithmic Trading, Alpha Research, Linear Programming

Analytics & AI: Regression, Time Series, Discrete Choice / Logit Models, Multi-Objective Optimization, CTGAN, LLM/RAG

Spoken Languages: English, Mandarin, Teochew, Malay